UNIT EIGHT

BURNS

COMPILED BY :

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DEFINITION:

Burn is an injury to body tissues caused by transfer of energy from heat source to the body. (heat, chemicals, or radiation).

✓Tissue destruction results from **coagulation, protein denaturation**, or **ionization** of cellular contents.

✓The depth of the injury depends on

- **temperature** of the burning agent &
- **duration of contact** with the agent.

BURNS ARE CAUSED MOST COMMONLY BY

- Carelessness
- Scalds
- Defective electrical equipment
- Use of open fires
- Immersion in an overheated bath water
- Use of chemicals such as strong acids

✓ TYPES OF BURNS (BASED ON CAUSE)

- **Dry Burn** – flames, lighted cigarettes, hot electrical equipments
- **Scalds burn**: steam, hot water or fat..
- **Cold burn**: – liquid oxygen(o2 that is cooled to – 187) & liquid nitrogen (damage the skin by freezing after prolonged exposure).
- **Chemical burn**: Acids & Alkalis
- **Electrical burn**: electric current & lightning
- **Radiation Burns**: Sunrays(UV light), x – ray

SYSTEMIC RESPONSE:-

- ✓ Loss of **capillary integrity**, leading **to increased capillary permeability** resulting in **massive fluid** losses/fluid shift from IC to EC.
- ✓Decrease CO, drop in BP
- ✓SNS increase peripheral resistance, increased in PR
- ✓Evaporative fluid loss through the burn wound may reach 3-5 liters/24hrs

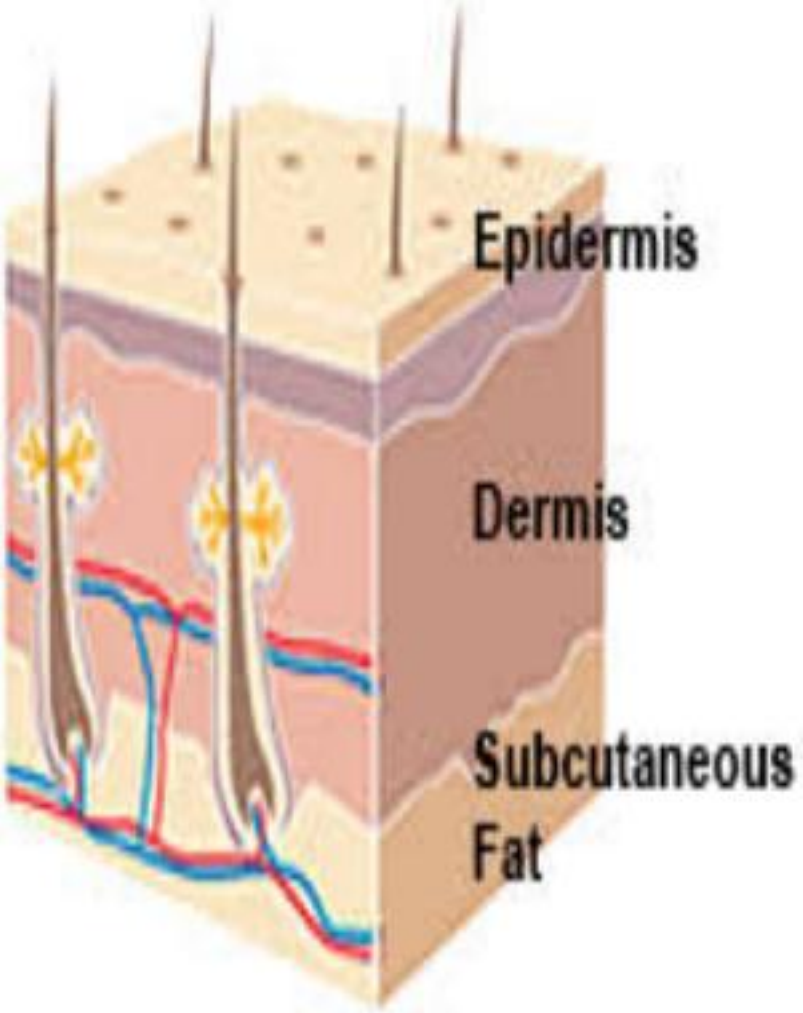
CLASSIFICATION

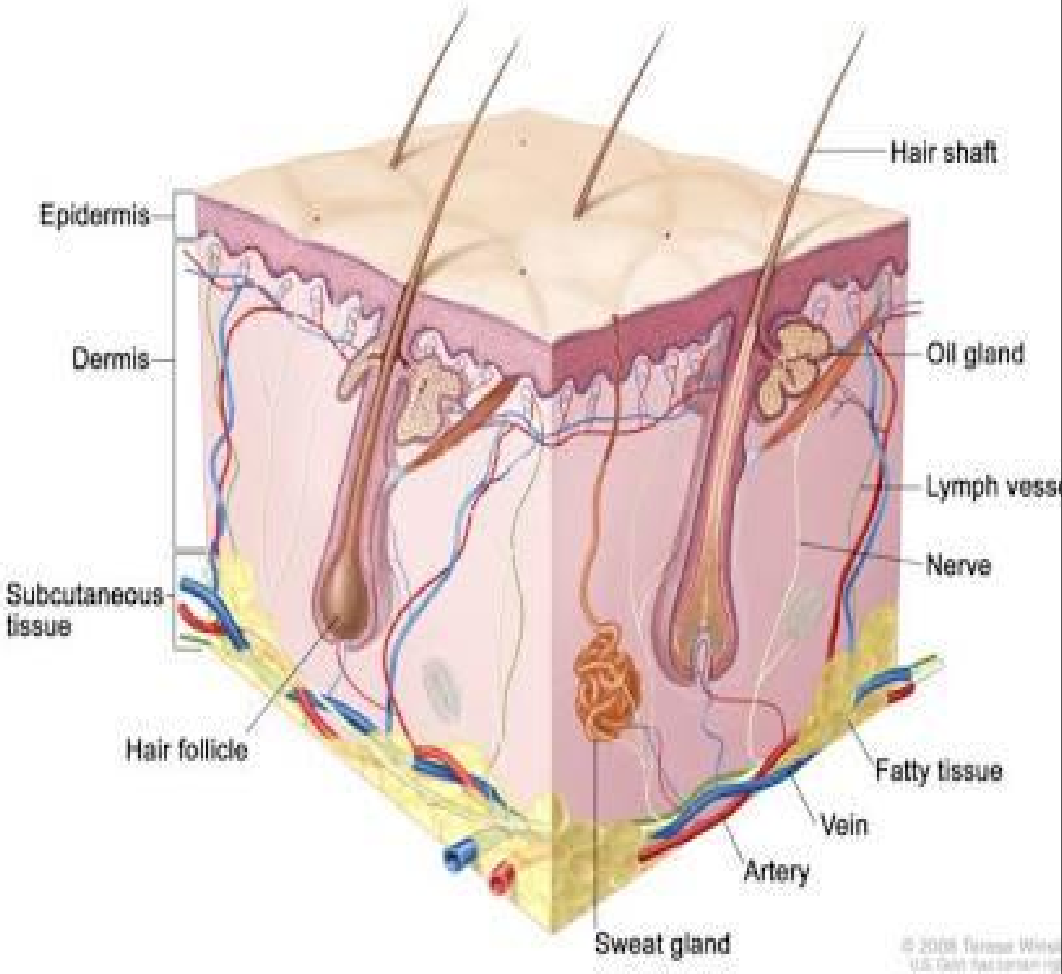
- Burns are usually classified according to depth or degree of skin damage

A. FIRST- DEGREE BURN(PARTIAL THICKNESS)

- ✓Limited to epidermis
- ✓Skin is fairy red & very pain full
- ✓No blisters
- ✓Heal in 3 – 7 days
- ✓No permanent scaring
  - ✓E.g. Sunburn

EPIDERMIS VS DERMIS VS HYPODERMIS





B. SECOND - DEGREE BURN (DEEP PARTIAL THICKNESS BURN)

- ✓Involve both epidermis & dermis
- ✓Skin is moist, pink or red
- ✓There are blisters
- ✓May or mayn't be painful
- ✓Takes weeks & leaves bad scar

C. THIRD DEGREE BURN (FULL THICKNESS BURN)

- Involves epidermis, dermis & subcutaneous tissues
- Skin charred & leathery or may be pale & dry
- Thrombosed veins are visible
- Pain is absent
- Healing occur only by skin grafting or scarring

D. FOURTH DEGREE BURNS

- Denotes involvement of muscles & bones

BURN

Burn classification

Characteristic	Superficial, partial-thickness (first-degree) burn	Deep, partial-thickness (second-degree) burn	Full-thickness (third-degree) burn
Appearance	Dry with no blisters	Weeping, edematous blisters	Dry, leathery, and possibly edematous
Color	Pink	White to pink or red	White to charred
Comfort	Painful	Very painful	Little or no pain
Depth	Epidermis only	Epidermis, dermis, and possibly some subcutaneous tissue	Subcutaneous tissue and possibly fascia, muscle, and bone

BURN

Table 57-1 • Characteristics of Burns According to Depth

DEPTH OF BURN AND CAUSES	SKIN INVOLVEMENT	SYMPTOMS	WOUND APPEARANCE	RECUPERATIVE COURSE
Superficial Partial-Thickness (Similar to First Degree)				
Sunburn Low-intensity flash	Epidermis; possibly a portion of dermis	Tingling Hyperesthesia (supersensitivity) Pain that is soothed by cooling	Reddened; blanches with pressure; dry Minimal or no edema Possible blisters	Complete recovery within a week; no scarring Peeling
Deep Partial-Thickness (Similar to Second Degree)				
Scalds Flash flame	Epidermis, upper dermis, portion of deeper dermis	Pain Hyperesthesia Sensitive to cold air	Blistered, mottled red base; broken epidermis; weeping surface Edema	Recovery in 2 to 4 weeks Some scarring and depigmentation contractures Infection may convert it to full thickness
Full-Thickness (Similar to Third Degree)				
Flame Prolonged exposure to hot liquids Electric current Chemical	Epidermis, entire dermis, and sometimes subcutaneous tissue; may involve connective tissue, muscle, and bone	Pain free Shock Hematuria (blood in the urine) and possibly hemolysis (blood cell destruction) Possible entrance and exit wounds (electrical burn)	Dry; pale white, leathery, or charred Broken skin with fat exposed Edema	Eschar sloughs Grafting necessary Scarring and loss of contour and function; contractures Loss of digits or extremity possible

ESTIMATING THE SEVERITY OF A BURN:-

There are different factors that combine to affect the severity of burn:

- ✓ **Size** : The larger the area of the burn, the more severe.
- ✓ **Cause:** The cause of the burn will influence the overall severity.

ESTIMATING THE SEVERITY OF A BURN:-

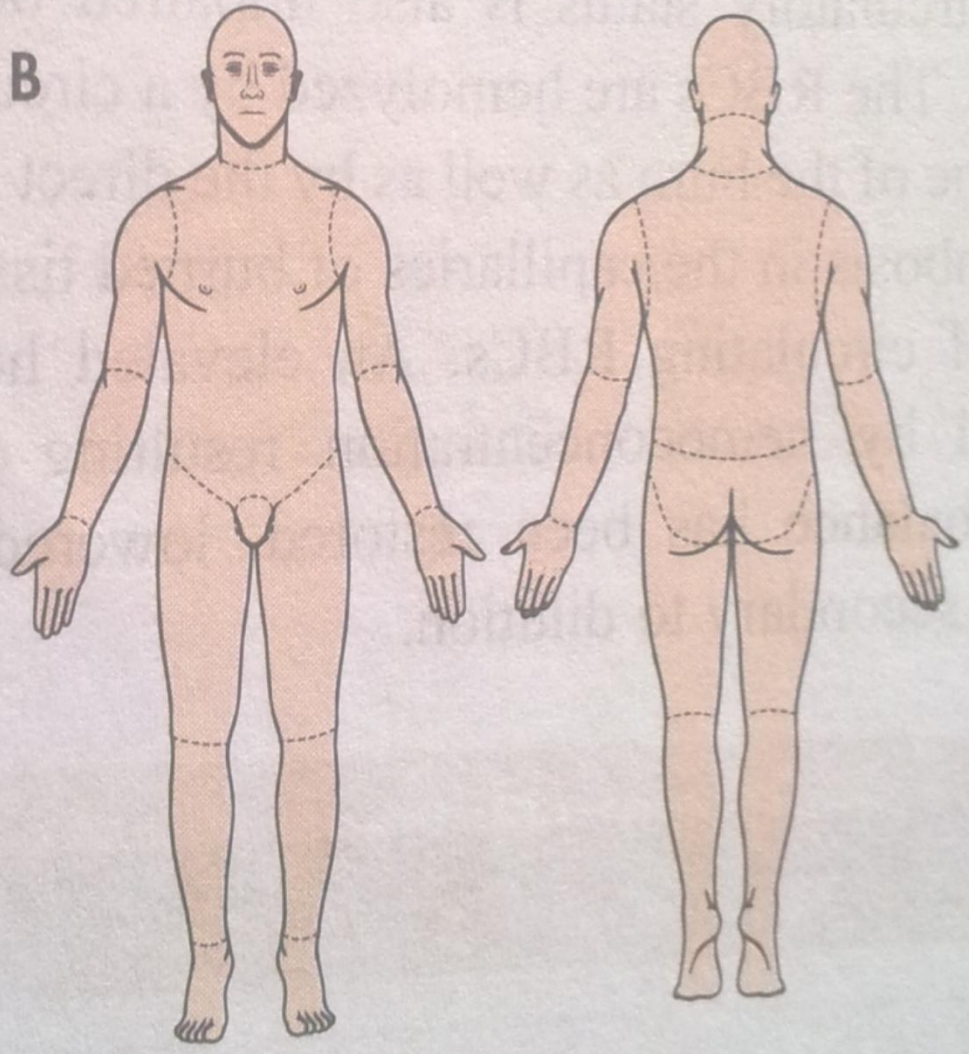
- ✓ **Age:** The age of the patient will affect the recovery rate and severity. Upper & lower lim
- ✓ **Location:** The location of the burn can affect the severity.(face)
- ✓ **Depth:** The deeper the burn, the more severe.

EXTENT & LOCATION OF BURNS

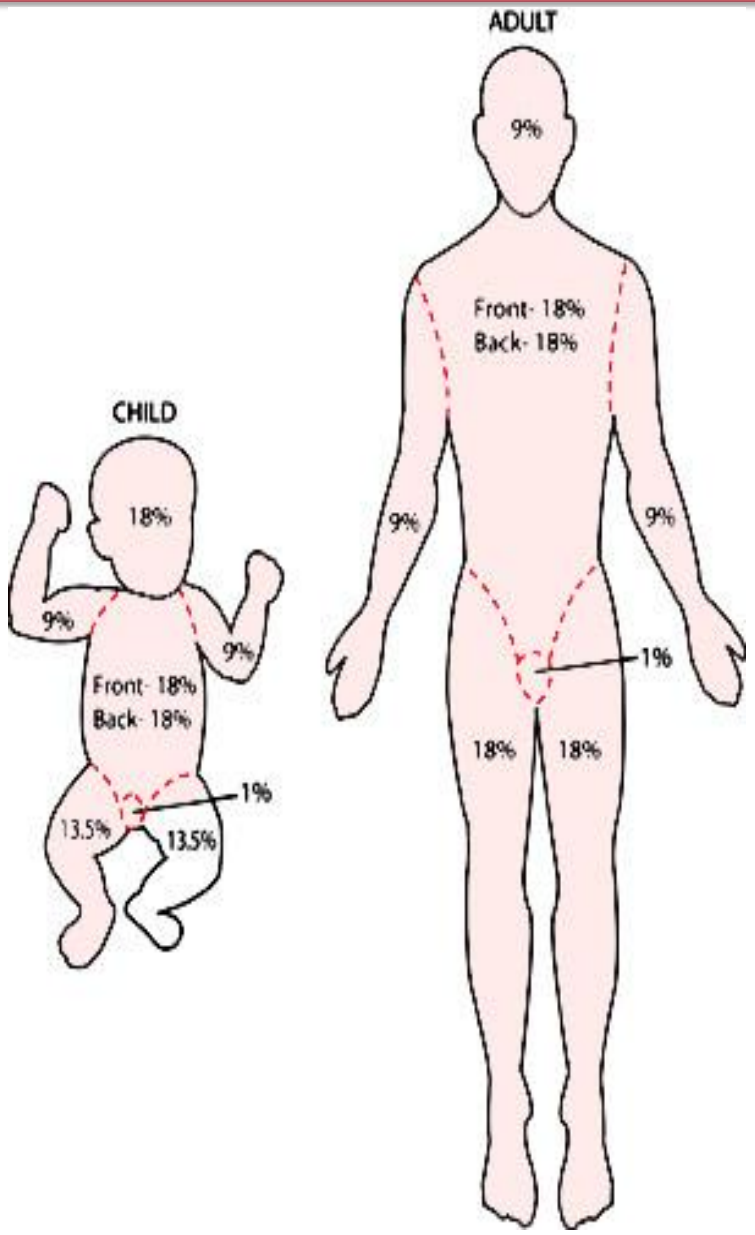
- The extent of the burn can be estimated by the percentage of body surface area involved (BSA)
- For burns that are more extensive, "the rule of nine" is used and for burns covering small area the "rule of palm" is used
- Rule of 9's for Adults: 9% for each arm, 18% for each leg, 9% for head,18% for front torso, 18% for back torso.
  Rule of 9's for Children: 9% for each arm, 14% for each leg, 18% for head, 18% for front torso, 18% for back torso

RULE OF NINE

Head & neck	9%
Arms (each)	9%
Ant. trunk	18%
Post. trunk	18%
Legs (each)	18%
Perineum	1%
TOTAL	100%



RULE OF NINE


ADULT
9%
Front- 18%
Back- 18%
9%
9%
1%
18%
18%
CHILD
18%
9%
9%
Front- 18%
Back- 18%
1%
13.5%
13.5%

SEEK MEDICAL ADVICE IF:-

- ✓The burn is larger than **1-inch square**.
- ✓ The patient is a **child.**
- ✓The burn goes **all the way around a limb**.
- ✓Any part of the burn appears to be **full thickness**.
- ✓The burn involves **hands, feet, genitals or the face**.
- ✓You are **not sure**.

EFFECTS

- Inhaling of poisonous gases like CO
- Asphyxia from insufficient oxygen
- Falls & injuries from collapsing

CRITICAL BURNS

- Burns of any degree involving more than 25% BSA
- 3rd degree burns involving more than 10% BSA
- Burns Complicated by respiratory injury
- Burns of the face, hands, feet or genitalia
- Burns in infants

FIRST AID MEASURES

AIM

To prevent further damage

To prevent infection

To rehabilitation

FIRST DEGREE BURN

- Extinguish flam by drop and roll
- Remove clothing
- Apply cold water or immerse the burned area in cool water
- Control pain
- Move to health facility

SECOND DEGREE BURN

- Immerse the burned part in cool water
- Elevate burned extremities
- Don't attempt to rupture blisters over the burn
- Control pain
- *Continuous observation for breathing difficulty*
- *Don't immerse an extensive burned area or apply ice over it because cold may **intensify** the shock reaction*
- *Arrange transportation to hospital quickly*
- ***NB** Don't use ointments, creams, sprays, or any similar material on any type of burn*

Above 4th Degree

- Open air way
- Wound care or dress the wound
- Elevate the affected part
- Treat for shock
- Refer the victim to nearby health institution





CONSENSUS FORMULA:-

- ✓ Lactated Ringer's solution (or other balanced saline solution): 2–4 ml x kg body weight x % TBSA burned.
- ✓ Half to be given in first 8 hours; half to be given over next 16 hours.
- ✓ e.g. – A 70-kg patient with a 50% TBSA burn
- ✓ First 8 hours = 3,500 ml; next 16 hours = 3,500 ml

EVANS FORMULA:-

Colloids:- 1 ml × kg body weight × %TBSA burned.

Electrolyte:- 1ml×body weight×% TBSA burned.

Glucose(5% In water):- 2000ml for insensible loss.

- ✓ Day1: Half given in first 8 hrs; remaining half over next 16 hrs
- ✓ Day2: Half of previous day's colloids and electrolytes.

Brooke Army formula:-

Colloids: 0.5 mL × kg body weight × % TBSA burned

Electrolytes (RL solution):-1.5mL × kg body weight × % TBSA burned

Glucose (5% in water):- 2,000 mL for insensible loss

Day 1: Half given in first 8 hrs; half over next 16 hrs.

Day 2: Half of colloids; half of electrolytes.

Parkland/Baxter Formula:- (commonly used)

- ✓ Lactated Ringer's solution: 4 ml × kg body weight × % TBSA burned.
- ✓ Day 1: Half to be given in first 8 hours; half to be given over next 16 hours
- ✓ Day 2: Varies. Colloid is added

General management of burn

- First Goal is to STOP THE BURNING PROCESS!

I. Stop, drop and roll. Smother with blanket or do use with water. DO NOT RUN!
II. Disconnect the person from the source of electricity
III. Remove clothing and jewelry. Take off blanket used to smother fire
IV. Cool burns or scalds by immediate immersion of water for at least 20 min.
V. Irrigation of chemical burns should be for 1 hour.
VI. Do NOT use ice for cooling
VII. Avoid hypothermia, keep the person as warm as possible.

NEVER

- ✓ NEVER burst blisters (the layer of skin is protecting against infection).
- ✓ NEVER touch the burn.
- ✓ NEVER apply lotions, ointments or fats – they might introduce infection.
- ✓ NEVER apply adhesive tape or dressings – the burn may be larger than it first appears.
- ✓ NEVER remove clothing that has stuck to the burn.

Effects of burn(contracture)








burn contracture Bitrus 4002.JPG

Thanks